PART-9

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(H) Graphical Representations :-

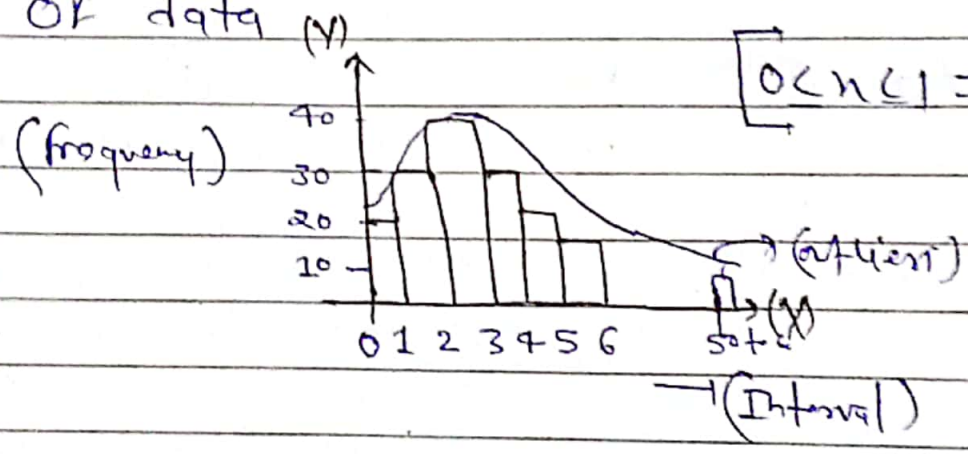
Three main plot are there

1. Histogram
2. Boxplot
3. Scatter plot

(I) Histogram :-

it displays the distribution of data

$\boxed{0 < n \leq 1 = 20}$



→ Histogram is also useful for identify pattern, outlier and skewness

(H) Type of skewed Histograms :-

1. symmetric (Normal distribution)
2. Right skewed

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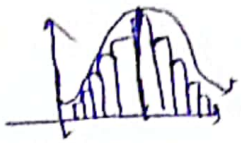
③ Left skewed

① Symmetric :-

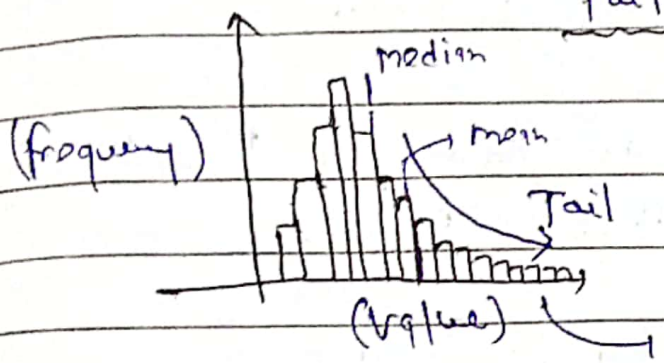


(Mean, Median, Mode)

(Center of distribution)

② Right skewed :-

Tail goes toward Right side



these all part is outliers

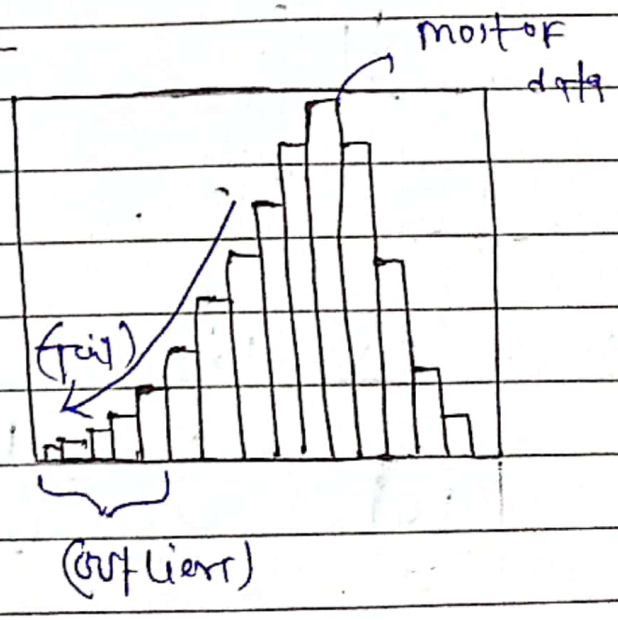
→ Mean - 2.05
median - 1.7

$\boxed{\text{Mean} > \text{median}}$

→ also called positive skewed

③ Left skewed :-

→ Tail goes toward left side



→ $\boxed{\text{median} > \text{mean}}$

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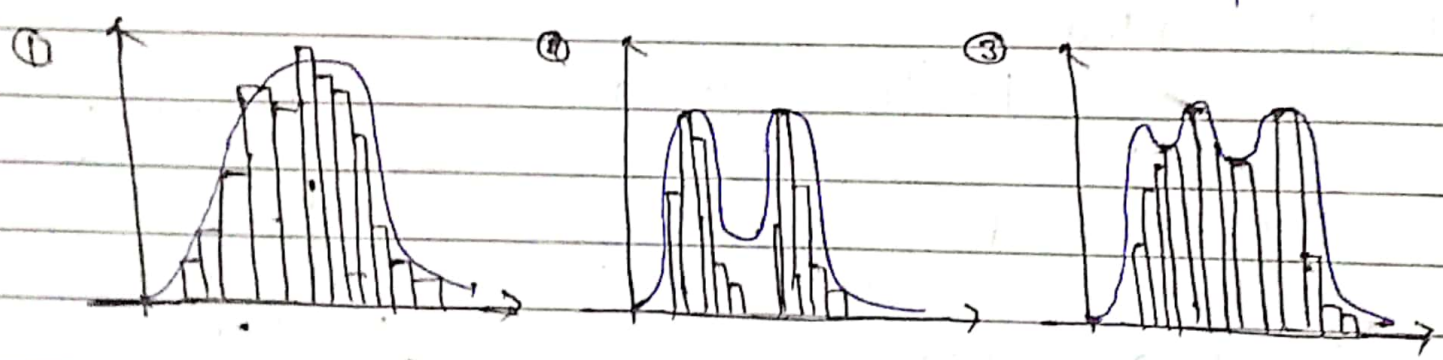
(ii) Types of Histogram based on number of modes:

① Unimodal Distribution Histo-
② Bimodal Distribution Histo-
③ Multimodal Distribution Histogram

① Unimodal Distribution somewhere it looks like Histogram with single peak.

② Bimodal Distribution have two peak.

③ Multimodal Distribution have multi peak.

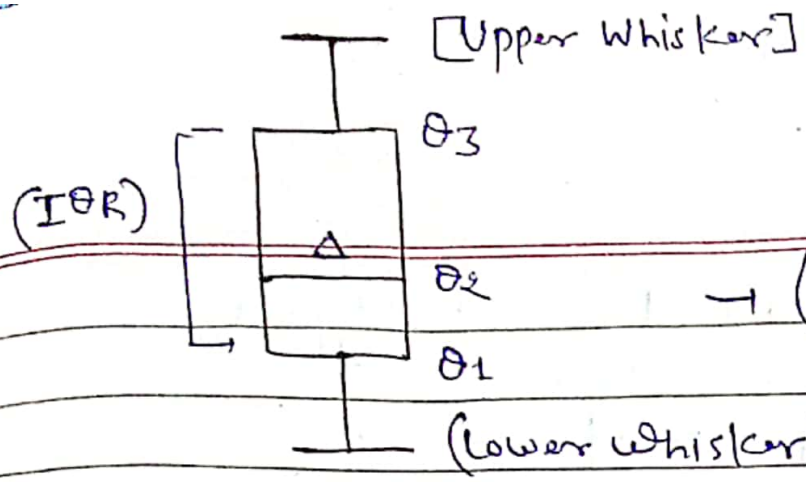


② Box plots:

Box-plot is nothing but

$$IQR = Q_3 - Q_1$$

→ it is useful to show the spread of data & identify outliers.



→ (it is also called Whisker plot)

(#) Upper Whisker = $Q_3 - (1.5 * IQR)$

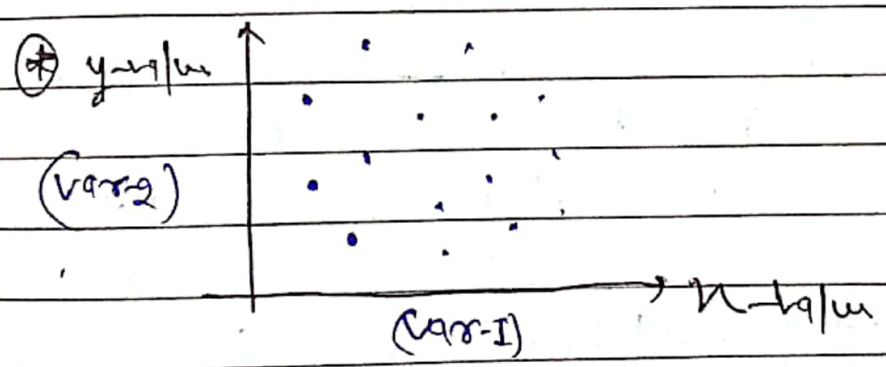
(#) Lower Whisker = $Q_1 - (1.5 * IQR)$

→ it is also called 5 Summary statistics.

→ it is used for identify the outliers

3. (#) Scatter plot :

Useful for visualizing the Relationship b/w two continous variable.



→ helps in identifying pattern, trends, and correlation.

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→ scatter plots useful for identifying outliers & understanding strength and direction of relation.

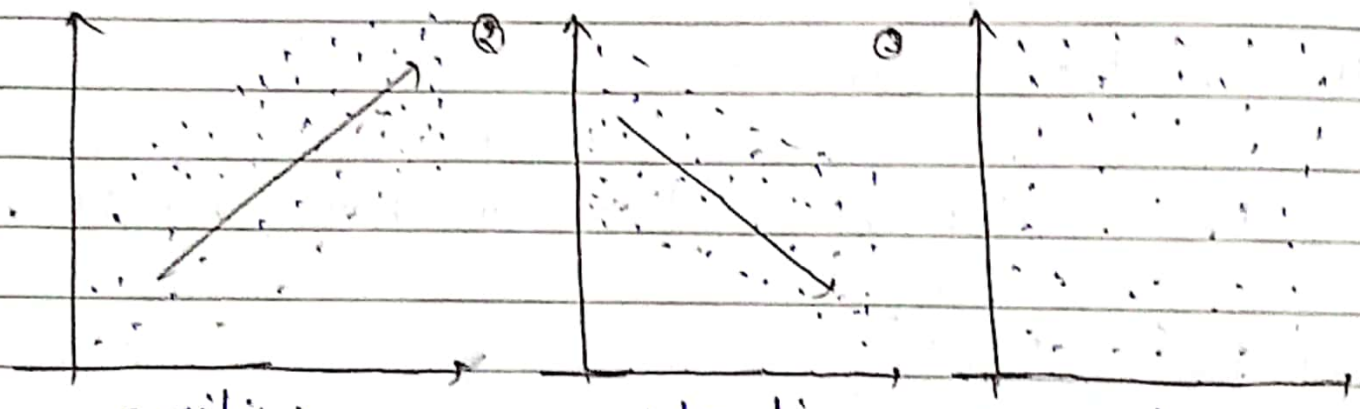
* Direction means:

① positive correlation
② Negative correlation
③ No correlation

* Relationship:

Relationship can be +ve, -ve and no relation.

①



positive correlation | Negative correlation | No correlation

→ var1(↑), var2(↑) → var1(↑), var2(↓)

→ var1 ∝ var2